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IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

Project ID: J26-SE-316

1. Topic (12 words max)

AI-Based Multi-Modal Web Threat Detection and Security Intelligence System

2. Research group the project belongs to

IAS - Information Assurance & Security

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature and Gravity of the Problem

Sri Lanka is experiencing a significant increase in phishing, fraudulent e-commerce, investment and cryptocurrency scams, brand impersonation, malicious websites, social-media fraud, and other web-based cyber threats alongside the country's rapid digital transformation.

Sri Lanka CERT received more than 12,650 complaints related to cybersecurity incidents and social-media misuse in 2025, demonstrating the growing scale of online fraud, hacked accounts, impersonation, harassment, and financial scams. In addition, the Kaspersky Security Bulletin reported 9,218 financial-phishing incidents in Sri Lanka during 2024, while Kaspersky products detected more than 8.6 million web-based cyber threats and 12.5 million local malware incidents on computers in Sri Lanka during the same period.

The social-media dimension is particularly important. Platforms such as Facebook, WhatsApp, Instagram, TikTok, and other online services are increasingly used to distribute scam advertisements, fake promotions, investment opportunities, job offers, cloned brand content, shortened malicious links, and links to fraudulent websites. These campaigns often combine multiple forms of deception rather than relying on a single malicious characteristic.

This creates an important research problem because many existing protection mechanisms analyze these signals independently. URL-based systems may identify suspicious lexical characteristics but fail to understand webpage content. Content-analysis systems may identify suspicious text but overlook brand impersonation in images. Visual-phishing systems may detect similar website screenshots but fail to correlate them with the original social-media post.

Affected parties and user personas (future users of the solution):

Persona 1 - Everyday Digital Consumer

A typical Sri Lankan internet user who uses social media, online shopping, digital banking, messaging platforms, and online services but may not have sufficient technical knowledge to distinguish legitimate websites and promotions from phishing pages, fraudulent stores, investment scams, or impersonated brands.

Persona 2 - Institutional Security / Fraud Professional

A security or fraud professional working in a bank, FinTech organization, e-commerce platform, telecommunications provider, SOC, or CERT who needs to investigate suspicious URLs, domains, webpages, social-media campaigns, and brand impersonation efficiently.

Persona 3 - Brand / E-Commerce Protection Team

Banks, e-commerce organizations, online brands, and digital-service providers are frequently impersonated using copied logos, website layouts, promotional images, fake checkout pages, and fraudulent advertisements.

Persona 4 - Developers and Code Owners

Software-development teams and organizations may unknowingly introduce vulnerabilities, insecure coding practices, hardcoded secrets, or insecure application logic that can later be exploited or reused in malicious infrastructure.

Impact of the Problem

1. Social Impact: Phishing, fake stores, investment scams, brand impersonation, and account takeovers reduce public trust in online banking, e-commerce, and digital services. Victims may suffer financial loss, identity theft, privacy violations, and psychological stress. ScamTrace helps by combining multiple threat indicators and providing clear explanations for suspicious websites and social-media content.

2. Technical Impact: Modern scams use newly registered domains, shortened URLs, redirects, multilingual scam content, deceptive HTML, obfuscated JavaScript, fake payment pages, cloned brands, and insecure code. These signals are often analyzed separately, which can miss complex attacks. ScamTrace combines URL/domain, webpage, behavioral, visual, social-media, and code-level analysis in one coordinated system.

3. Economic Impact: Online scams cause direct financial losses through credential theft, fraudulent payments, and fake investment schemes. Organizations also face costs from incident response, refunds, chargebacks, account recovery, and reputational damage. ScamTrace aims to reduce these losses through earlier detection and actionable threat intelligence.

4. Ethical and Legal Impact: Phishing, identity theft, unauthorized access, and personal-data theft create serious privacy and legal concerns. These activities may relate to Sri Lanka's Computer Crime Act No. 24 of 2007 and Personal Data Protection Act No. 9 of 2022. ScamTrace supports safer digital services through explainable detection and human-controlled remediation.

5. Environmental Impact: Automated threat analysis can reduce repetitive manual investigation and improve incident-response efficiency. Secure digital services can also support less paper-based and travel-intensive activity. ScamTrace uses a layered approach so lightweight analysis can be performed before more computationally expensive AI and visual processing.

FoC Research Cluster and Industry Verticals

1. Research Cluster: The primary research cluster is Information Assurance & Security (IAS), covering web threat detection, phishing and scam detection, URL/domain intelligence, behavioral intent analysis, brand impersonation, vulnerability detection, and cyber threat intelligence. The project also aligns with Artificial Intelligence & Data Science through machine learning, multilingual NLP, computer vision, explainable AI, and LLM-assisted remediation.

2. Industry Verticals: ScamTrace is relevant to Digital Banking and FinTech, E-Commerce, Telecommunications, Cybersecurity/SOCs, Software Development and DevSecOps, and Public Sector/CNII. These sectors can use the system for phishing detection, scam analysis, brand protection, threat intelligence, and secure software auditing.

Contribution to Sustainable Development Goals

SDG 8 - Decent Work and Economic Growth: ScamTrace can contribute to protecting household finances, SME revenue, online businesses, digital banking, and e-commerce from cyber-enabled fraud. Reducing online financial scams can help maintain confidence in the digital economy and support technological upgrading and digital economic activity under SDG Target 8.2.

SDG 9 - Industry, Innovation and Infrastructure: ScamTrace combines machine learning, multilingual NLP, computer vision, threat intelligence, static program analysis, and LLM-assisted remediation to strengthen secure digital infrastructure. The project therefore supports innovation and the development of resilient and secure ICT infrastructure, particularly within financial, commercial, and government digital services.

SDG 16 - Peace, Justice and Strong Institutions: Automated identification and explanation of phishing, scams, malicious infrastructure, impersonation campaigns, and software vulnerabilities can support CERTs, cybersecurity teams, financial institutions, and investigative organizations.

References:

[1] Sri Lanka Computer Emergency Readiness Team (CERT|CC), *Information and Cyber Security Strategy of Sri Lanka (2019–2023)*, 2018. [Online]. Available: https://www.cert.gov.lk/wp-content/uploads/policies/Information_and_CyberSecurity_StrategyofSri-Lanka.pdf

[2] Sri Lanka Computer Emergency Readiness Team (CERT|CC), *National Cyber Security Strategy of Sri Lanka (2025–2029)*, 2025. [Online]. Available: https://www.cert.gov.lk/wp-content/uploads/policies/National_Cyber_Security_Strategy_of_Sri-Lanka.pdf

[3] United Nations, *Sustainable Development Goals (SDGs).* [Online]. Available: https://sdgs.un.org/goals

6. Brief description of Existing Research and Systems (300 words max)

Existing products/systems: Current commercial defenses largely depend on URL blacklists, reputation services, browser warnings, and endpoint-security products such as Google Safe Browsing, PhishTank, OpenPhish, and URLhaus. These systems are effective for known malicious infrastructure but are mainly reactive because suspicious URLs or domains normally need to be discovered or reported before blocking. Social-media platforms rely on keyword filters, automated moderation, and user reporting, while banks and e-commerce platforms use separate fraud-detection rules. Visual brand-protection tools can identify copied logos or impersonated websites, but are usually separated from URL/domain and webpage-behavior analysis. Similarly, SAST and repository scanners identify vulnerable code independently from live web-threat intelligence and often focus on findings rather than context-aware remediation.

Existing research: Research on web-threat detection has advanced across several specialized areas, but these are still commonly investigated independently. URL and domain research uses lexical characteristics, DNS/WHOIS data, domain age, DGA detection, redirects, and machine-learning classifiers to identify malicious infrastructure before webpage execution. Webpage studies analyze HTML structure, visible text, forms, JavaScript, and multilingual content to detect phishing, fraudulent e-commerce, investment scams, and other deceptive behavior. Sánchez-Paniagua et al. (2021), for example, combined HTML, textual, and visual features for fake-store detection, while later research applied transformer-based language models to investment and cryptocurrency scams. Visual-phishing systems such as VisualPhishNet and Phishpedia showed that screenshots, logos, and visual similarity can identify brand impersonation. Separately, software-security research combines AST-based analysis, SAST tools, vulnerability taxonomies, and LLMs to generate vulnerability explanations and candidate patches. These studies show that URL intelligence, webpage content, visual evidence, social-media signals, and code weaknesses are individually detectable, but rarely unified.

Gaps/limitations identified: (1) URL/domain, webpage-content, visual-phishing, social-media scam, and source-code security research are mostly developed as separate pipelines, limiting cross-source evidence correlation. **(2)** URL/domain detectors can identify suspicious lexical, DNS, registration, DGA, newly registered domain, shortened-link, and redirect characteristics, but cannot explain the malicious content or visual deception shown after the link is opened. **(3)** HTML and text-based detectors can identify suspicious forms, multilingual scam language, or page structure, but often do not jointly analyze static JavaScript intent such as form hijacking, clipboard manipulation, deceptive redirects, or data-flow indicators. **(4)** Social-media and visual-phishing research can detect reused images, logos, or visually similar webpages, but commonly analyzes the post and destination independently instead of correlating the social-media "bait" with the linked "trap." **(5)** Brand-similarity techniques are rarely combined with URL/domain intelligence, OCR text, logo detection, linked-site screenshots, and cross-source consistency scoring. **(6)** Traditional SAST and repository-auditing systems identify vulnerabilities but remain disconnected from broader threat-analysis platforms and often stop at findings rather than context-grounded remediation. **(7)** No single lightweight architecture integrates intelligent URL/domain analysis, tri-modal HTML/content/static behavioral-intent analysis, social-media and visual brand analysis, and proactive repository auditing with LLM-assisted human-reviewed remediation. This is the key gap addressed by **ScamTrace**.

7. Brief description of the solution's nature, including a conceptual diagram.

Brief Description of the Solution

The proposed solution is **ScamTrace: AI-Based Multi-Modal Web Threat Detection and Security Intelligence System**, an integrated framework that discovers and analyzes suspicious web threats using four complementary research components. **ThreatRadar** supports the discovery of suspicious social-media posts, URLs, and newly registered domains. **C1 - Intelligent URL & Domain Analyzer** performs lexical, structural, DNS, registration, DGA, short-link, and redirect analysis. **C2 - Tri-Modal HTML, Content & Behavioral Intent Analyzer** examines DOM structure, multilingual webpage content, and static JavaScript intent without requiring heavyweight sandbox execution. **C3 - Social Media & Web Visual Threat Analyzer** correlates social-media text, images, OCR evidence, logos, brand similarity, and linked webpage screenshots to identify visual impersonation and social-to-web inconsistencies. In parallel, **C4 - Code Quality & Vulnerability Auditor with LLM-Assisted Remediation** performs proactive repository security analysis using AST/SAST techniques and generates context-grounded remediation suggestions for human review. The outputs of C1–C3 are combined into an explainable web-threat assessment, while C4 provides a complementary upstream software-security workflow.

How the Solution Addresses the Identified Gaps

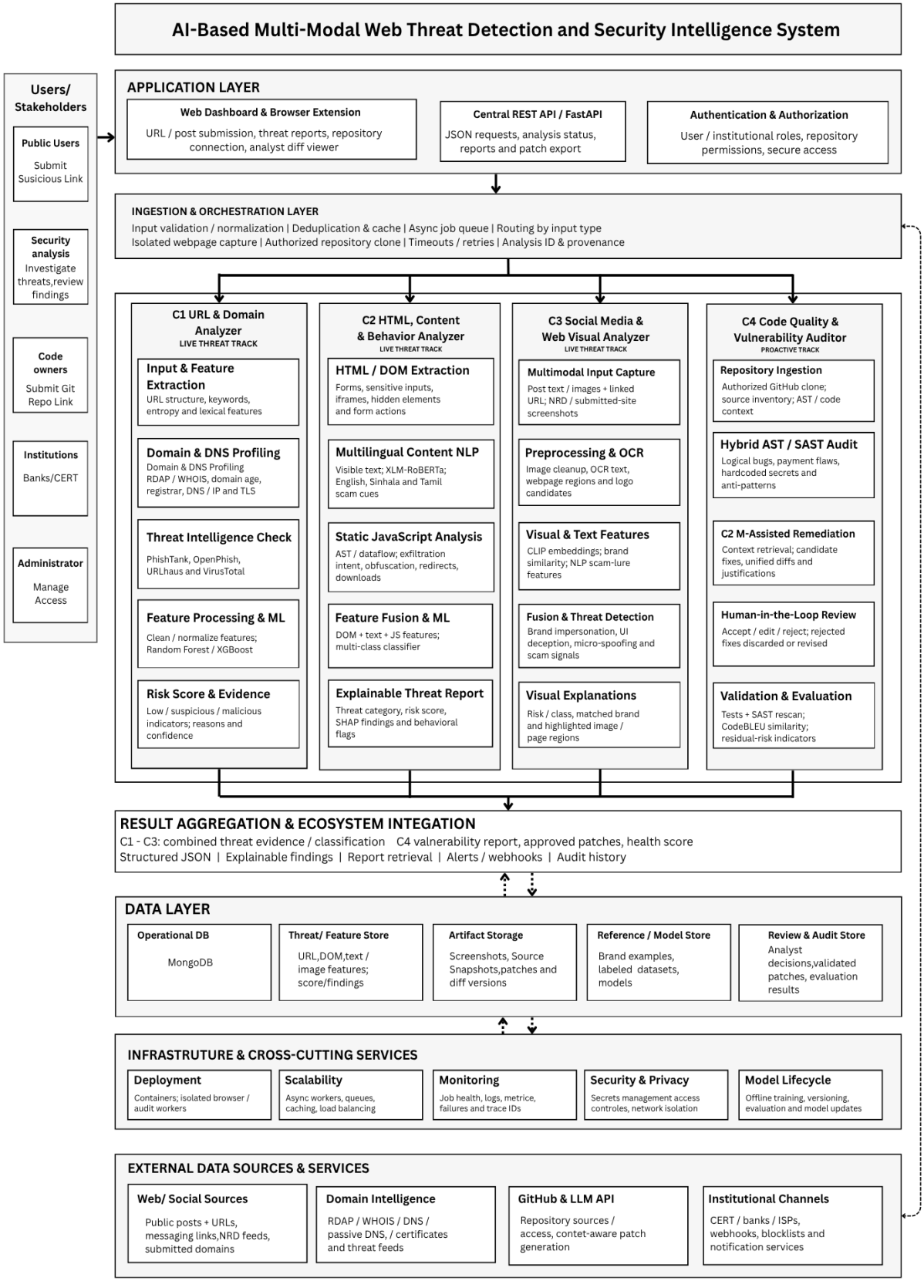
Each gap identified in Section 6 is addressed through a specific ScamTrace design decision. **(1) Fragmented threat-analysis pipelines** are addressed by integrating URL/domain, webpage, social-media, and visual evidence within a common architecture rather than using separate tools for each signal. **(2) Limitations of URL/domain-only detection** are addressed by allowing C1 findings to be complemented by C2 webpage-content and behavioral-intent evidence, providing information about both suspicious infrastructure and what the destination page attempts to do. **(3) Limited analysis of HTML and textual content** is addressed by C2, which combines DOM structural features, multilingual NLP, and static JavaScript/dataflow analysis to detect indicators such as deceptive forms, hidden elements, redirects, clipboard manipulation, and form-hijacking intent without depending on expensive dynamic sandbox execution. **(4) Isolation between social-media scams and destination websites** is addressed by C3, which analyzes the original social post and follows its URL to the linked webpage, correlating the social-media "bait" with the destination "trap." **(5) Limited visual brand verification** is addressed through OCR, CLIP-based brand similarity, logo/UI detection, linked-site screenshot comparison, and cross-source consistency analysis to detect copied branding and mismatched destination domains. **(6) Separation between web-threat detection and software-security auditing** is addressed by C4, which identifies repository vulnerabilities using AST/SAST analysis and generates context-aware LLM-assisted patches that remain subject to human approval. **(7) Lack of a unified explainable architecture** is addressed through standardized component outputs and evidence fusion, enabling ScamTrace to generate threat scores, confidence values, categories, and reasons rather than only a binary safe/unsafe verdict.

System Perspective (Input – Process – Output)

Input: ScamTrace accepts URLs, domains, newly registered domains, social-media posts containing text, images and links, and user-submitted suspicious websites. The system also uses webpage artifacts such as HTML, DOM structure, JavaScript, visible text and destination screenshots, together with reference information including legitimate brand logos, official domains, known malicious URLs, scam samples, and labeled training datasets. For C4, the input is a public or authorized private GitHub repository URL containing web-application source code to be audited.

Process: (i) ThreatRadar discovers and prioritizes suspicious social-media posts, URLs, and domains for analysis. (ii) C1 validates and normalizes URLs, extracts lexical and structural features, analyzes DNS/RDAP/WHOIS and registration information, evaluates newly registered domains and DGA characteristics, and investigates shortened URLs and redirect chains. (iii) C2 processes HTML, DOM, visible text and JavaScript to extract structural features, multilingual scam-language indicators, and static behavioral-intent evidence such as suspicious forms, data flows, redirects, clipboard manipulation, or form hijacking. (iv) C3 preprocesses social and webpage images, performs OCR, identifies logos and brand candidates, calculates CLIP-based visual similarity, captures linked webpages, and measures consistency between the social post, claimed brand, URL, and destination website. (v) A fusion layer combines C1–C3 evidence into an overall web-threat assessment with granular category scores, confidence, and explainable reasons. (vi) Separately, C4 retrieves repository source code, performs AST/SAST and code-quality analysis, records weaknesses with severity and CWE evidence, generates context-grounded remediation suggestions and unified diffs, and presents them for human-in-the-loop acceptance, rejection, or modification.

Output: For runtime web analysis, ScamTrace produces an overall threat score, confidence level, threat-category assessment, and explainable evidence report containing URL/domain indicators, webpage-content findings, static behavioral-intent evidence, social-media signals, OCR results, brand-impersonation evidence, visual similarity, and cross-source inconsistencies. The results are made available through the ScamTrace API and dashboard for users, financial institutions, e-commerce organizations, security teams, and CERT personnel. For C4, the system produces a repository-security report containing vulnerability and code-quality findings, severity and exploitability information, CWE references, proposed LLM-assisted patches, unified diffs, and the final human-review decision.


AI-Based Multi-Modal Web Threat Detection and Security Intelligence System
Users/ Stakeholders
Public Users
Submit Susicious Link
Security analysis
Investigate threats,review findings
Code owners
Submit Git Repo Link
Institutions
Banks/CERT
Administrator
Manage Access
APPLICATION LAYER
Web Dashboard & Browser Extension
URL / post submission, threat reports, repository connection, analyst diff viewer
Central REST API / FastAPI
JSON requests, analysis status, reports and patch export
Authentication & Authorization
User / institutional roles, repository permissions, secure access
INGESTION & ORCHESTRATION LAYER
Input validation / normalization | Deduplication & cache | Async job queue | Routing by input type
Isolated webpage capture | Authorized repository clone | Timeouts / retries | Analysis ID & provenance
C1 URL & Domain Analyzer
LIVE THREAT TRACK
Input & Feature Extraction
URL structure, keywords, entropy and lexical features
Domain & DNS Profiling
Domain & DNS Profiling
RDAP / WHOIS, domain age, registrar, DNS / IP and TLS
Threat Intelligence Check
PhishTank, OpenPhish, URLhaus and VirusTotal
Feature Processing & ML
Clean / normalize features; Random Forest / XGBoost
Risk Score & Evidence
Low / suspicious / malicious indicators; reasons and confidence
C2 HTML, Content & Behavior Analyzer
LIVE THREAT TRACK
HTML / DOM Extraction
Forms, sensitive inputs, iframes, hidden elements and form actions
Multilingual Content NLP
Visible text; XLM-RoBERTa; English, Sinhala and Tamil scam cues
Static JavaScript Analysis
AST / dataflow; exfiltration intent, obfuscation, redirects, downloads
Feature Fusion & ML
DOM + text + JS features; multi-class classifier
Explainable Threat Report
Threat category, risk score, SHAP findings and behavioral flags
C3 Social Media & Web Visual Analyzer
LIVE THREAT TRACK
Multimodal Input Capture
Post text / images + linked URL; NRD / submitted-site screenshots
Preprocessing & OCR
Image cleanup, OCR text, webpage regions and logo candidates
Visual & Text Features
CLIP embeddings; brand similarity; NLP scam-lure features
Fusion & Threat Detection
Brand impersonation, UI deception, micro-spoofing and scam signals
Visual Explanations
Risk / class, matched brand and highlighted image / page regions
C4 Code Quality & Vulnerability Auditor
PROACTIVE TRACK
Repository Ingestion
Authorized GitHub clone; source inventory; AST / code context
Hybrid AST / SAST Audit
Logical bugs, payment flaws, hardcoded secrets and anti-patterns
C2 M-Assisted Remediation
Context retrieval; candidate fixes, unified diffs and justifications
Human-in-the-Loop Review
Accept / edit / reject; rejected fixes discarded or revised
Validation & Evaluation
Tests + SAST rescan; CodeBLEU similarity; residual-risk indicators
RESULT AGGREGATION & ECOSYSTEM INTEGATION
C1 - C3: combined threat evidence / classification C4 valnerability report, approved patches, health score
Structured JSON | Explainable findings | Report retrieval | Alerts / webhooks | Audit history
DATA LAYER
Operational DB
MongoDB
Threat/ Feature Store
URL,DOM,text / image features; score/findings
Artifact Storage
Screenshots, Source Snapshots,patches and diff versions
Reference / Model Store
Brand examples, labeled datasets, models
Review & Audit Store
Analyst decisions,validated patches, evaluation results
INFRASTRUTURE & CROSS-CUTTING SERVICES
Deployment
Containers; isolated browser / audit workers
Scalability
Async workers, queues, caching, load balancing
Monitoring
Job health, logs, metrice, failures and trace IDs
Security & Privacy
Secrets management access controles, network isolation
Model Lifecycle
Offline training, versioning, evaluation and model updates
EXTERNAL DATA SOURCES & SERVICES
Web/ Social Sources
Public posts + URLs, messaging links,NRD feeds, submitted domains
Domain Intelligence
RDAP / WHOIS / DNS / passive DNS, / certificates and threat feeds
GitHub & LLM API
Repository sources / access, contet-aware patch generation
Institutional Channels
CERT / banks / ISPs, webhooks, blocklists and notification services

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Gaining Specialized Domain Expertise.

The project sits within the Information Assurance & Security domain, with supporting expertise in AI/ML, computer vision, NLP, threat intelligence, and secure software engineering. Team members will strengthen their domain knowledge through guidance from the academic supervisor and co-supervisor, consultation with the Faculty's security research community, and, where possible, feedback from practitioners such as Sri Lanka CERT, banking-sector security teams, and software-security professionals. The team will also study literature related to URL/domain analysis, multilingual scam detection, static JavaScript intent analysis, visual phishing and brand impersonation, and LLM-assisted vulnerability remediation. This will help each member understand both established and recent approaches before designing, implementing, and evaluating the four ScamTrace components.

Gaining Technical Knowledge and Skills.

Each member owns a separate technical area aligned with the four research components: intelligent URL/domain analysis, tri-modal HTML/content/static behavioral-intent analysis, social-media and visual threat analysis, and code-quality/vulnerability auditing with LLM-assisted remediation. Required skills include Python, FastAPI, machine learning, XLM-RoBERTa and multilingual NLP, DOM and JavaScript parsing, Tree-sitter and Semgrep, CLIP, YOLO-based logo detection, Siamese/visual-similarity models, OCR, explainable AI, PostgreSQL, and REST API integration. These skills will be developed through official documentation, peer-reviewed research, reputable online courses, and incremental hands-on prototyping. Reproducing baseline models and comparing individual modalities before fusion will provide practical understanding while also establishing benchmarks for evaluation. Regular team code reviews, integration sessions, and supervisor checkpoints will support technical knowledge transfer across the group.

Data Collection

The project uses a combination of public benchmark datasets, live threat feeds, legitimate reference data, and self-collected samples. Malicious URLs and domains will be obtained from sources such as PhishTank, OpenPhish, URLhaus, and other public threat-intelligence feeds, while benign domains will be collected from sources such as Tranco. URL/domain datasets will include lexical, DNS, RDAP/WHOIS, newly registered domain, DGA, short-link, and redirect information where available. For C2, webpage HTML, DOM structure, JavaScript, visible multilingual text, and scam-lure samples will be collected for static content and behavioral-intent analysis. For C3, public datasets such as VisualPhish, Phish30k, Phishpedia, and other screenshot/logo datasets will be combined with legitimate brand logos, official domains, social-media posts, OCR text, and linked-page screenshots collected by the project. For C4, public or authorized GitHub repositories and known vulnerable-code examples will be used for AST/SAST evaluation and remediation testing. Collected data will be cleaned, deduplicated, labeled, and divided into training, validation, and test sets, while avoiding dependence on restricted third-party sources.

Ethical Clearance

The project does not involve human participants, personal data collection, surveys, or interviews as part of its core methodology, and therefore does not require full ethical-review-board clearance for human-subject research. However, because it handles live malicious content and phishing pages that may reference real brands and (incidentally) personal data, the team will observe responsible-research practices: malicious samples are executed only within isolated sandboxes, no live phishing infrastructure is attacked or interfered with, harvested artifacts are anonymized where they contain personal identifiers, and all handling complies with the Personal Data Protection Act No. 9 of 2022 and the Computer Crime Act No. 24 of 2007. If any human-facing evaluation (e.g., user testing) is later introduced, appropriate ethical clearance will be obtained beforehand.

9. Objectives and Novelty

Main Objective

To design and develop ScamTrace: AI-Based Multi-Modal Web Threat Detection and Security Intelligence System that proactively discovers and analyzes suspicious web threats from social media platforms, user submissions, and newly registered domains by combining intelligent URL/domain analysis, tri-modal HTML/content/static behavioral-intent analysis, social-media and web visual threat analysis, and proactive code-quality/vulnerability auditing with LLM-assisted remediation. The system aims to detect and explain phishing, fraudulent e-commerce, financial and investment scams, malware-related threats, brand impersonation, data-theft attempts, and other deceptive web activities by correlating multiple evidence sources rather than relying on blacklists or single-modality detection. It further provides explainable threat scores, category-level results, supporting evidence, and human-reviewed remediation, enabling more accurate and practical security intelligence for users, organizations, and security teams.

Member Name	Sub Objective	Tasks	Novelty
M.S.N. Peiris - IT23201132	Develop a Code Quality & Vulnerability Auditor with LLM-Assisted Remediation that analyzes public or authorized web-application repositories using static program analysis to identify security vulnerabilities, insecure coding patterns, logical weaknesses, and code-quality issues, and generates context-aware remediation suggestions without	• Develop GitHub repository ingestion and secure source-code collection. • Perform AST-based code analysis using Tree-sitter and static vulnerability detection using Semgrep/custom security rules. • Identify vulnerabilities such as insecure authentication, injection risks, hardcoded secrets, unsafe input handling, insecure dependencies, and application-logic weaknesses.	This component introduces a hybrid neuro-symbolic vulnerability auditing framework that combines deterministic AST/SAST analysis with context-grounded LLM-assisted remediation. Unlike conventional scanners that mainly report vulnerabilities, the proposed approach connects each verified finding to a targeted code context and produces an auditable unified patch for human review. A key novelty is the Human-in-the-Loop remediation workflow, which prevents unsafe autonomous code modification while allowing

	executing untrusted repository code.	• Map findings to severity levels, CWE categories, affected files, and supporting code evidence. • Build a context-grounding layer that selects relevant code and vulnerability evidence for the LLM remediation engine. • Generate security-fix suggestions and unified code diffs using an LLM. • Implement a Human-in-the-Loop (HITL) review workflow where users can accept, reject, or manually modify suggested patches. • Calculate a repository-level Infrastructure Health & Exploitability Score and generate an explainable security report.	developers to accept, reject, or edit proposed fixes. The component also introduces an Infrastructure Health & Exploitability Score to summarize repository-level security risk. By analyzing repositories statically without executing untrusted source code, it provides proactive and safer software-security analysis that complements ScamTrace's runtime web-threat detection components.
M.T.C. Peiris - IT23201200	Develop a Social Media & Web Visual Threat Analyzer that detects and explains scam campaigns by jointly analyzing social-media post text, images, embedded URLs, OCR-extracted text, brand/logo evidence, and the visual appearance of linked destination webpages.	• Collect and preprocess social-media posts containing text, images, and URLs together with linked webpage screenshots. • Validate, resize, normalize, deduplicate, and prepare images for visual analysis. • Apply multilingual OCR to extract embedded text and store OCR confidence values.	This component introduces a cross-source social-to-web visual threat intelligence framework that analyzes the complete scam path from the social-media “bait” to the linked webpage “trap.” Unlike conventional screenshot-similarity or logo-detection systems that analyze a webpage in isolation, the proposed approach correlates

	The component identifies brand impersonation, deceptive promotional content, suspicious UI patterns, and inconsistencies between the claimed brand, social post, destination domain, and linked webpage.	• Develop CLIP-based zero-shot brand detection to identify likely brand impersonation, including previously unseen visual variations. • Apply YOLO-based logo/UI detection to identify logos, payment elements, login forms, buttons, and suspicious interface regions. • Use visual-similarity/Siamese methods to detect logo and UI micro-spoofing against legitimate brand references. • Capture the destination webpage linked from the social-media post and compare its visual identity with the post and legitimate brand references. • Calculate cross-source consistency between post text, detected brand, visual evidence, destination domain, and linked webpage. • Produce granular scores for brand impersonation, visual similarity, image/text threat evidence, and overall social-visual risk.	social-post text, OCR evidence, image content, detected brand/logo, destination domain, and linked-site visual identity. It combines CLIP-based zero-shot brand recognition, YOLO-based logo/UI detection, multi-granular visual similarity, micro-spoofing analysis, and explainable visual evidence. Another novelty is the use of cross-source consistency scoring, which can detect cases where a post visually claims to represent a trusted brand but redirects users to an unrelated or suspicious destination. The component produces multiple explainable threat scores rather than only a single similarity value, supporting detection of brand impersonation, phishing-style UI deception, financial-scam interfaces, and other visually driven social-media scams.

		• Apply XAI techniques such as Grad-CAM/attention maps to highlight suspicious regions and generate human-readable reasons for the final threat report.	
W A D A K Appuhamy - IT23479746	Develop an Intelligent URL & Domain Analyzer capable of detecting phishing URLs, malware-distribution domains, suspicious newly registered domains, shortened malicious links, DGA-generated domains, homoglyph/obfuscated URLs, and malicious redirect infrastructure using lexical, structural, DNS, registration, and machine-learning-based analysis without requiring webpage execution. The component provides lightweight pre-execution screening and explainable risk evidence for the wider ScamTrace platform.	• Validate, normalize, and parse submitted URLs and domains. • Extract lexical URL features such as URL length, special characters, suspicious keywords, numeric patterns, character repetition, and entropy. • Analyze structural characteristics including subdomain depth, path complexity, query structure, encoding, homoglyphs, and URL obfuscation. • Detect Domain Generation Algorithm (DGA) patterns using statistical, entropy-based, and ML-based features. • Collect domain intelligence using RDAP/WHOIS, registration age, registrar information, DNS records, nameservers, IP/ASN information, and TLS/reputation indicators where available. • Identify suspicious Newly	This component proposes a Multi-Layer URL & Domain Intelligence Framework that combines lexical and structural URL analysis, domain-registration intelligence, DNS/infrastructure evidence, DGA detection, NRD risk analysis, homoglyph and obfuscation detection, short-link resolution, and redirect-chain analysis within one lightweight pre-execution pipeline. Unlike traditional blacklist-based solutions that depend heavily on previously reported malicious URLs, the proposed approach attempts to identify suspicious or previously unseen infrastructure from intrinsic URL and domain characteristics. A key novelty is the generation of granular, explainable evidence showing which URL, registration, DNS, redirect, or DGA indicators contributed to the final

		Registered Domains (NRDs) and abnormal registration characteristics. • Detect shortened URLs and safely resolve their hidden destinations. • Analyze redirect chains to identify suspicious multi-hop redirection, destination changes, and malicious redirect infrastructure. • Engineer and preprocess URL/domain features and remove duplicate samples from datasets. • Train and compare baseline models such as Decision Tree, Random Forest, XGBoost, and LightGBM. • Evaluate using Accuracy, Precision, Recall, F1-Score, ROC-AUC, False Positive Rate, and latency. • Generate an explainable URL/Domain Risk Score and structured evidence for integration with the ScamTrace threat-fusion layer.	risk score. The component is designed as ScamTrace's early screening layer, allowing suspicious URLs and domains to be prioritized before more expensive webpage-content and visual analysis by C2 and C3.

Dilshan.N - IT23250574	Develop a Tri-Modal HTML, Content & Behavioral Intent Analysis component that examines webpage DOM structure, multilingual textual content, and static JavaScript/dataflow intent to detect multi-class web threats such as phishing, fake e-commerce, investment scams, malware-delivery, and browser abuse without requiring dynamic page execution. The component combines structural, semantic, and behavioral-intent evidence into one explainable threat assessment.	• Extract and parse HTML/DOM structure, including forms, inputs, scripts, iframes, hidden elements, and external resources. • Identify credential-harvesting forms, fake payment interfaces, deceptive checkout elements, and investment-scam indicators. • Apply multilingual NLP using XLM-RoBERTa to detect phishing/scam language in English, Sinhala, and Tamil. • Build and annotate a multilingual scam-lure corpus for training and evaluation. • Detect malware-delivery indicators such as fake software-update buttons and suspicious download links. • Perform static JavaScript intent analysis for form hijacking, clipboard manipulation, forced redirects, hidden navigation, and deceptive interaction logic. • Perform static dataflow analysis to trace sensitive form submissions, external endpoints, and suspicious script execution paths. • Extract DOM structural features and multilingual text embeddings and combine them with static	This component introduces a tri-modal, cross-lingual, structure-aware threat analysis framework that jointly analyzes DOM structure, multilingual scam semantics, and static JavaScript behavioral intent. Unlike approaches that rely only on webpage text, only DOM features, or dynamic sandbox execution, the proposed component infers malicious intent directly from static webpage artifacts. A key novelty is the combination of XLM-RoBERTa-based multilingual semantic analysis with DOM structural features and static JavaScript/dataflow indicators, enabling detection of suspicious behaviors such as external credential submission, form hijacking, clipboard manipulation, deceptive redirects, and malware-delivery logic without executing the page. The component also produces granular, explainable evidence showing which DOM, textual, and JavaScript indicators contributed to the final

		JavaScript behavioral-intent features. • Perform feature engineering, cleaning, labeling, and dataset preprocessing for multi-class classification. • Train and compare models such as XGBoost and Random Forest for threat classification. • Evaluate using Precision, Recall, F1-Score, confusion matrices, false-positive rate, and processing latency. • Conduct an ablation study comparing DOM-only, NLP-only, behavioral-intent-only, and tri-modal fusion configurations. • Generate structured JSON containing threat category, risk score, confidence, and explainable DOM/text/JavaScript evidence for ScamTrace integration.	classification, supporting phishing, fake e-commerce, investment-scam, malware-delivery, and browser-abuse detection within one lightweight pipeline.

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IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

10. This section is to be completed by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Data required for the project is available.

Yes	✓	No	

f) Any other comments:

–

	Title	First Name	Last Name	Signature
Supervisor*	Mr.	Eishan	Weerasinghe	
Co-Supervisor*	M.S.	Madusha	Weerasooriya	(On behalf of Madusha)
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.